Problem Set 4

Starter

1 (E)**.** Let G be a finite group of order n with a subgroup H of order $n-1$. What can you say about n?

2 (E)**.** Show that if a group G contains an element of order six, and an element of order ten, then G has order at least 30.

3 (E)**.** Let G be a finite group.

(a) Let $a \in G$. Show that there is a positive integer n such that $a^n = e$, the identity element.

(b) Show that there exists a positive integer n such that $a^n = e$ for all $a \in G$.

4 (L-M)**.** Let K, H be subgroups of a group G with $K \leq H \leq G$. Show that $[G : K] = [G : H][H : K]$. If any two of these indices are finite, then so is the third.

Advanced

5 (L-M)**.** Let H be a subgroup of the group G. Find a (natural) bijection between the set of all left cosets and the set of all right cosets of H in G.

6 (M)**.** Let G be a finite group and H a subgroup of G.

(i) Let x be an element of order d in G. If r is the least positive integer such that $x^r \in H$, show that r divides d.

(ii) Suppose further that H has index n. If $x \in G$, show that $x^k \in H$ for some k such that $0 < k \leq n$. Is it always the case that the least positive such k is a factor of n? Justify your answer.

7 (M)**.** Let $\mathbb{C}^*$ be the multiplicative group of nonzero complex numbers. Suppose that H is a subgroup of finite index of $\mathbb{C}^*$. Prove that $H = \mathbb{C}^*$.

8 (H)**.** Let G be finite group such that there exists $g \in G$ with more than $3/4$ of the elements x of G satisfying $x^2 = g$. Show that every nonidentity element of G has order 2.